Matthias **König**

Digital Twins · AI · Digital Pathology · Pharmacometrics · Open & FAIR

University Hospital Schleswig-Holstein, Campus Lübeck, First Department of Medicine

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Carrer Stages & Education

Professor of Metabolic Inflammation and Carcinogegensis of the Liver — Lübeck University, Germany

Systems Medicine and Digital Twins — 06/2026 -

- Lead the **Systems Medicine and Digital Twins** at the University of Lübeck. Development of digital twins, digital pathology methods.

Associate Professor of Pharmacoepidemiology and Data Science — SDU, Odense, Denmark

Pharmacoepidemiology and Data Science — 05/2026

- Lead the **Pharmacoepidemiology and Data Science Group** at the University of Southern Denmark (SDU), advancing research in real-world data analytics, causal inference, and translational data science.

Head of Computational Biomechanics Group — University of Stuttgart

Coordination Project, DFG Priority Programme SPP 2311 — 12/2025 - 04/2026

- Lead the **Computational Biomechanics Group** at the University of Stuttgart.
- Coordinate interdisciplinary biomechanics and numerical modeling for robust, translational in silico models.
- Integrate modeling, simulation, and software usability toward clinical applications.
- Promote community building, standardization, and reproducibility within the SPP.

Head of Systems Medicine of the Liver Group — Humboldt-Universität zu Berlin

DFG Position for Principal Investigators | Systems Medicine of the Liver — 06/2021 – Present

- Established and lead the **Systems Medicine of the Liver Group** at the Institute for Theoretical Biology.
- Secured competitive third-party funding (DFG, EU, BMBF) to advance digital physiology and personalized pharmacotherapy.
- Developed innovative digital twin models integrating physiology, pharmacokinetics, and AI-based decision support.
- Supervised and mentored PhD, Master, and Bachelor students, fostering interdisciplinary and open-science skills.
- Built international collaborations (Virtual Human Twin, EDITH, Lorentz Center) to strengthen systems medicine research.

Junior Group Leader — Humboldt-Universität zu Berlin

BMBF LiSyM Junior Group Leader | Systems Medicine of the Liver — 2016 – 2021

- Founded an independent junior research group focused on multiscale computational modeling of liver physiology and disease.
- Developed predictive PBPK/PD models for drug metabolism, liver surgery outcomes, and metabolic disorders.
- Contributed to national flagship program **LiSyM – Liver Systems Medicine** with interdisciplinary and translational projects.
- Initiated and coordinated collaborative projects linking clinical hepatology, surgery, and computational biology.

PhD in Biophysics — Charité – University Medicine Berlin, Humboldt-Universität zu Berlin

Doctorate in Biophysics (magna cum laude) | Computational Modeling of Glucose Metabolism — 2010 - 2015

- Researched and developed detailed kinetic models of hepatic glucose metabolism and whole-body regulation.
- Demonstrated mechanisms underlying glucose homeostasis and hypoglycemia risk in type 2 diabetes.
- Supervisor: Prof. Hermann-Georg Holzhütter (Charité Berlin).

Scientific Staff Member — Charité – University Medicine Berlin

Researcher in HepatoSys and Virtual Liver initiatives — 2008 – 2015

- Contributed to two major BMBF-funded consortia (**HepatoSys**, **Virtual Liver**) advancing systems medicine of the liver.
- Built multi-scale models of hepatic metabolism, perfusion, and drug elimination, bridging basic research and clinical application.
- Collaborated closely with experimental and clinical partners to validate models with in vitro, in vivo, and patient data.

Diploma in Biophysics — Humboldt-Universität zu Berlin

Diploma (equivalent to MSc, 1.0 with distinction) — 2002 – 2009

- Acquired a strong foundation in biophysics, physiology, and computational modeling.
- Diploma thesis on systems biology modeling of hepatic metabolism, laying the groundwork for doctoral research.

Surgical Support Assistant (Rotational Role) — Bethesda Hospital Stuttgart

Clinical experience prior to academic studies 2001

- Assisted surgical teams with preparation, organization, and post-operative support across multiple departments.
- Coordinated with nurses, anesthesiologists, and surgeons to maintain high standards of patient safety and workflow efficiency.
- Gained early exposure to interdisciplinary teamwork in high-pressure clinical environments.

Training & Certifications

Leadership Certificate Program — ZEWK, TU Berlin

Certificate in Academic Leadership (100 AE) — 2025

- Completed a two-stage leadership program designed for junior professors and early-career group leaders.
- Trained in collaborative and respectful teamwork, effective communication, and conflict resolution.
- Strengthened ability to lead within collegial academic structures, balancing autonomy and institutional expectations.
- Enhanced professional skills to take on leadership roles with clarity and confidence.

Digital Health Professions Educator (DHPE) — Charité – Universitätsmedizin Berlin

Certificate in Digital Teaching for Health Professions (200 AE) — 2024 – 2025

- Completed faculty development program focused on digital teaching and innovation in healthcare education.
- Designed and implemented future-oriented teaching scenarios, integrating blended and digital learning.
- Enhanced institutional teaching capacity through innovative, research-based learning formats.
- Applied competencies to improve curriculum development and digitally supported learning in health sciences.

AI Teaching Certificate (KI-Lehrzertifikat) — HU Berlin, KI-Campus 2.0 & AI Skills Initiative

Higher education didactics qualification on Artificial Intelligence (200 AE) — 2025

- Completed: AI Foundations (96%), AI Advanced & Ethics (93%), and AI Didactics in Higher Education (91%).
- Covered core AI topics: machine learning, neural networks, Python/Jupyter, text and image processing, and responsible AI use.
- Training aligned with DGHD standards, including ethics, data and algorithmic governance, and regulatory aspects of AI.
- Focus on AI-supported teaching: generative AI, speech assistants, prompt design, development of AI-based OER teaching units.
- Included a teaching observation and reflective assessments demonstrating competence in designing and implementing AI-enhanced learning scenarios.

Science Communication Certificate Program — ZEWK, TU Berlin

Certificate in Science Communication (120/200 AE) — 2025 - 2026

- Gained advanced training in effectively communicating research to diverse audiences including the public, media, funding bodies, and scientific peers.
- Acquired skills in media and press work, strategic communication, and legal frameworks in science communication.
- Developed practical competencies in writing, presenting, and visual design through project-based exercises.
- Applied knowledge in real-world communication formats to enhance outreach and visibility of scientific work.

EP PerMed Training on Scientific Integrity — European Partnership for Personalised Medicine

Training on Research Integrity and ELSA (20 AE) — 2025

- Participated in international training on ethical, legal, and social aspects (ELSA) of personalised medicine.
- Studied European Code of Conduct guidelines and key principles of research integrity.
- Learned procedures for good research practice, ethics compliance, and responsible dissemination of results.
- Strengthened understanding of integrity issues in biomedical and translational research.

IAR Identify, Assist, Refer — Health & Wellness, University of Toronto

Certificate in Mental Health Assistance and Support (10 AE) — 2025

- Completed training on identifying mental health challenges and offering appropriate assistance.
- Gained skills in facilitating help-seeking behavior and providing mental health support in challenging situations.
- Learned strategies for effective communication and crisis intervention.
- Acquired practical knowledge to support individuals experiencing mental health difficulties.

Skills

Languages: **German**, **English**, Spanish (B2)

Teaching & Communication: **University Teaching**, Project-Based Learning, Digital Learning Tools, Science Communication, Mentoring & Supervision, Public Engagement

Research & Modeling: **Physiologically Based Pharmacokinetic/ Pharmacodynamic (PBPK/PD)**, Systems Biology, Multiscale Modeling, Machine Learning & Artificial Intelligence, FAIR Data Management, Reproducible Workflows, Computational Simulation, Digital Pathology

Programming **Python**, **R**, **Julia**, Java, C++, MATLAB, SQL, HTML, JavaScript, PyTorch, TensorFlow, SBML

Tools & Software **git**, **PyCharm**, **VS Code**, **Excel**, LaTeX, Typst, Word, PowerPoint, Canva, BioRender, Adobe Illustrator, Inkscape

1. **Parameter Uncertainty in the Optimization of Pharmacokinetic Models: A Reproducible Bayesian Approach**. Antonio Alavarez (supervisor: **Matthias König**); Master Thesis, April 2026, Last Author
2. **A Systems Pharmacology Approach to Rivaroxaban: Physiologically Based Modeling of Pharmacokinetics and Coagulation Dynamics**. Elisabetta Casabianca (supervisor: **Matthias König**); Master Thesis, July 2025, Last Author
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13. **Computational modelling of omeprazole - pharmacokinetics and pharmacodynamics**. Sükrü Balci (supervisor: **Matthias König**); Bachelor Thesis, October 2021, Last Author
14. **A Physiologically Based Model of Indocyanine Green Liver Function Tests - Effects of Physiological Factors, Hepatic Disease and Hepatic Surgery**. Adrian Köller (supervisor: **Matthias König**); Bachelor Thesis, March 2021, Last Author
15. **Computational Modelling of Simvastatin - Effects on HMG-CoA Reductase Activity and Cholesterol**. Florian Bartsch (supervisor: **Matthias König**); Bachelor Thesis, November 2020, Last Author
16. **Computational Modelling of Midazolam Clearance: Effect of Inhibitors and Inducers**. Yannick Duport (supervisor: **Matthias König**); Bachelor Thesis, August 2020, Last Author

Scientific Results

Ten Selected Publications (total: 66, HF: 26:, i10: 36, citations: 3778, Google Scholar 08/26)

1. **Automated Segmentation of Hepatic Vessels and Lobules in Whole-Slide Images Using U-Net Models**. M. Bafna, M. König, S. Saalfeld, V. Moulisova, V. Liska, U. Dahmen, and M. Albadry; Front. Bioinform. 6:1713736, 10.3389/fbinf.2026.1713736, IF: **3.9**
2. **A Physiologically Based Pharmacokinetic and Pharmacodynamic (PBPK/PD) Model of Dapagliflozin in Type 2 Diabetes Mellitus: The Effect of Dosing, Hepatorenal Impairment, and Food**. N. Nemitz, M. Elias, and M. König; Pharmaceutics 2026, 18(3), 287; https://doi.org/10.3390/pharmaceutics 18030287, 10.3390/pharmaceutics18030287, IF: **6.9**, Last author
3. **A Digital Twin of Glimepiride for Personalized and Stratified Diabetes Treatment**. M. Elias and M. König; Front. Pharmacol. 16:1686415., 10.3389/fphar.2025.1686415, IF: **4.8**, Last author
4. **Cross-Species Variability in Lobular Geometry and Cytochrome P450 Hepatic Zonation: Insights into CYP1A2, CYP2E1, CYP2D6 and CYP3A4**. M. Albadry, J. Kuettner, J. Grzegorzewski, O. Dirsch, E. Kindler, R. Klopfleisch, V. Liska, V. Moulisova, S. Nickel, R. Palek, J. Rosendorf, S. Saalfeld, U. Settmacher, H. Tautenhahn, M. KönigΔ, and U. DahmenΔ (Δ equal contribution); Front Pharmacol. 2024 May 16;15:1404938, 10.3389/fphar.2024.1404938, IF: **5.4**, Last Equal author
5. **A pathway model of glucose-stimulated insulin secretion in the pancreatic β-cell**. M. D. Maheshvare, S. Raha, M. KönigΔ, and D. PalΔ (Δ equal contribution); Front. Endocrinol. 14:1185656, 10.3389/fendo.2023.1185656, IF: **5.7**, Last Equal author
6. **Physiologically based pharmacokinetic (PBPK) modeling of the role of CYP2D6 polymorphism for metabolic phenotyping with dextromethorphan**. J. Grzegorzewski, J. Brandhorst, and M. König; Front Pharmacol. 2022 Oct 24;13:1029073, 10.3389/fphar.2022.1029073, IF: **4.4**, Last author
7. **Prediction of survival after hepatectomy using a physiologically based pharmacokinetic model of indocyanine green liver function tests**. A. Köller, J. Grzegorzewski, M. Tautenhahn, and M. König; Front. Physiol., 22 November 2021, 10.3389/fphys.2021.730418, IF: **3.2**, Last author
8. **PK-DB: pharmacokinetics database for individualized and stratified computational modeling**. J. Grzegorzewski, J. Brandhorst, K. Green, D. Eleftheriadou, Y. Duport, F. Barthorscht, A. Köller, D. Y. J. Ke, S. De Angelis, and M. König; Nucleic Acids Res. 2021 Jan 8;49(D1):D1358-D1364, 10.1093/nar/gkaa990, IF: **16.7**, Last author
9. **HEPATOKIN1: A Biochemistry-Based Model of Liver Metabolism Suited for Applications in Medicine and Pharmacology**. N. Berndt, S. Bulik, I. Wallach, T. Wünsch, M. König, M. Stockmann, M. Meierhofer, and H. G. Holzhütter; Nat Commun. 2018 Jun 19;9(1):2386., 10.1038/s41467-018-04720-9, IF: **15.4**
10. **Quantifying the Contribution of the Liver to the Homeostasis of Plasma Glucose: A Detailed Kinetic Model of Hepatic Glucose Metabolism Integrated with the Hormonal Control by Insulin, Glucagon and Epinephrine**. M. König, S. Bulik, and H. G. Holzhütter; PLoS Comput Biol. 2012 Jun;8(6):e1002577. Epub 2012 Jun 21., 10.1371/journal.pcbi.1002577, IF: **4.9**, First author

For a full list of publications and preprints see: https://livermetabolism.com/publications/.

Ten Selected Research Outcomes

1. **An Open and Reproducible Digital Twin for Personalised Anticoagulant Therapy: Modelling Apixaban Pharmacokinetics and Pharmacodynamics**. M. Myshkina, M. Babaeva, L. C. Nguyen, S. Bedón, J. Metternich, M. Elias, and M. König; Preprints 2026, 10.20944/preprints202605.0249.v1, 10.20944/preprints202605.0249.v1, Last author
2. **Quantitative Image Analysis of Hepatic Zonation in Cytochrome P450 and Steatosis Using Whole Slide Scans**. Master Thesis Jonas Küttner (supervisor: **Matthias König**); Master Thesis, August 2024, Last Author
3. Preprint **Assessing the Impact of AI and Digital Twins on Clinical Decision-Making in Hepatology and Hepatobiliary Surgery**. Mariia Myshkina, Elisabetta Casabianca, Anton Schnurpel, Tim Ricken, Hans-Michael Tautenhahn, **Matthias König**; Preprints 2025, 10.20944/preprints202509.1164.v1, Last Author
4. Database **PK-DB - Pharmacokinetics database**. doi:10.1093/nar/gkaa990 Developed the first FAIR-compliant open database for pharmacokinetics, integrating clinical and pre-clinical trial data. PK-DB enables reproducible PBPK/PD modeling, supports individualized simulations, and has become a key infrastructure for computational pharmacology research.

5. Software **sbmlutils - Python utilities for SBML**. doi:10.5281/zenodo.597149 Created a versatile Python library to streamline the use of SBML models, providing robust utilities for model handling, analysis, and integration with libSBML. Widely used in reproducible modeling workflows across systems biology.
6. Software **SBML4Humans - SBML simulation made easy**. Designed an interactive reporting framework that makes SBML models human-readable and accessible, enabling experts and newcomers to explore model content without technical barriers.
7. Software **sbmlsim - SBML simulation made easy**. Built a lightweight Python package that simplifies simulations of SBML models on top of libRoadRunner, lowering the entry barrier for model testing and teaching.
8. Software **cysbml - Cytoscape 3 app for the Systems Biology Markup Language**. doi:10.5281/zenodo.597154 Developed and maintained a widely used Cytoscape app for visualization of SBML models in network contexts. cy3sbml has facilitated intuitive exploration of complex models in systems biology and bioinformatics.
9. Software **libsbgnpy - Python library for SBGN**. Developed a Python library for working with Systems Biology Graphical Notation (SBGN), supporting standardized visualization and integration of pathway information.
10. Software **roadrunner - High-performance simulator for SBML**. Contributed to the development of libRoadRunner, a C/C++ library using LLVM for ultra-fast simulation of SBML models, setting a benchmark for performance in computational biology.

Supervision of Researchers

Over the last 5 years, 1 PhD theses, 4 Master thesis, 11 Bachelor's theses, 18 Humboldt-Internship students (HIC), and 4 Google Summer of Code (GSOC) students have been supervised. I am currently supervising 2 PhD, and 2 Internship projects (group members).

1. **Parameter Uncertainty in the Optimization of Pharmacokinetic Models: A Reproducible Bayesian Approach**. Antonio Alvarez (supervisor: M. König); Master Thesis, April 2026, Supervisor
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6. **A Physiologically-Based Pharmacokinetic Model of Tirzepatide**. Internship Report Abhinav Mishra (supervisor: M. König); Internship Report, Mar 2025, Supervisor
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11. A physiologically-based pharmacokinetic (PBPK) model of sorafenib for investigating the effect of sorafenib parameters in hepatorenal impairment. Internship Report Frances Okibedi (supervisor: M. König); Internship report, July 2024, Supervisor

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Activities in the Research System

- 2026: **Invited keynote at the Patient Monitoring Roundtable (PMRT)** – Digital Twins and Dashboards for Personalized Drug Treatment at Charité – Universitätsmedizin Berlin.
- 2026: **Interview: Modelle so komplex wie der Körper selbst – Digitale Zwillinge in der Medizin** – Gesellschaft für Informatik interview on digital twins, multi-scale modeling, PBPK, and open, FAIR computational medicine.
- 2025: **Featured in Faces of the Berlin University Alliance** – Highlighted as part of the BUA's 6th-anniversary film series showcasing interdisciplinary collaboration across FU, HU, TU, and Charité.
- 2025–2026: **Leading X-Student Research Group: Digital Twins in Action — Optimizing Direct Oral Anticoagulant Use** - Supervising students in PBPK modeling and advancing precision medicine.
- 11/2025: **Organizing Workshop Open Science & Reproducibility — For Computational Models in Systems Biology & Medicine** - Featuring interactive sessions (LEGO® Serious Play), reproducibility workflows, and open publishing practices.
- 2025–2026: **Member Research Data Alliance (RDA) Working Group: Building Immune Digital Twins** - Developing standards and methods for Digital Twins.
- 2018–2023, 2025–2027: **Elected SBML Editor** - Advanced the Systems Biology Markup Language (SBML) standard by coordinating editorial processes, improving specifications, and fostering interoperability.
- 2024–2026: **Open Science Ambassador, Humboldt-University Berlin** - Promoted reproducibility, FAIR data, and open science practices within the Berlin University Alliance.
- 2017–2022, 2024–2027: **Elected SED-ML Editor** - Enhanced the Simulation Experiment Description Markup Language to improve reproducibility and exchangeability of simulation experiments.
- 2023–2026: **Elected PEtab Editor** - Strengthened the PEtab standard for parameter estimation problems, ensuring community adoption and long-term sustainability.
- 2017–2026: **Elected COMBINE Coordinator** - Led international coordination of community standards (SBML, SED-ML, PEtab, etc.), strengthened collaborations, and organized annual COMBINE meetings.
- 2025: **Interview: Reproducibility, Open Science, and the Future of Biological Research** - Berlin University Alliance feature highlighting the reproducibility crisis, FAIR data, open standards, and cultural change in academia.

- 2023: **Led X-Student Research Group: Physiologically Based Digital Twins for Hypertension Therapy** - Focused on ACE inhibitors and diuretics.
- 2023–2024: **Expert Panel Member: PharmVar CYP1A2 Gene Panel** - Advanced pharmacogenomics standards in drug metabolism.
- 2023: **Co-authored Open Science Concept: Eleven Strategies for Training Reproducible Research** - Implemented institutional strategies for reproducible research and open science training.
- 2022: **Organized 13th Computational Modeling in Biology Network (COMBINE) Meeting** - Brought together global experts in modeling standards in Berlin.
- 2022: **Participant Lorentz Workshop "Your Dietary Digital Twin"** – Contributed to discussions on integrating mechanistic and data-driven models for personalized nutrition and open science frameworks.
- 2022: **Led X-Student Research Group: Physiologically Based Modeling of Drugs — ACE Inhibitors in Hypertension** - Mentored interdisciplinary student teams.

Funding

- 10/2025 – 04/2026, Recipient, BMBF, 1.500€, **X-Student Research Group - Digital Twins in Action: Optimizing Direct Oral Anticoagulant Use.** Project-based teaching initiative on digital twins in medicine, focusing on physiologically based pharmacokinetic (PBPK) modeling to optimize the use of direct oral anticoagulants.
- 10/2025 – 09/2026, Recipient, DFG, 15.000€, **SPP2311 - Startup Funding - SPP-FEMVis: Advancing Open Science with Web-Based FEM Visualization.** Startup funding to develop an open science platform for interactive, web-based visualization of finite element method (FEM) simulations.
- 2024 – 2025, Co-Investigator, Circle U., total: 10.000€, **AlgoNomy - Algorithmic Regulation Before Medical Liability - Advancing Doctor-Patient Autonomy in AI-Driven Healthcare.** Building an international academic network to investigate the role of algorithms and AI in medical decision support systems, with a focus on legal, ethical, and patient autonomy implications.
- 01/2024 – 06/2026, Recipient, BMBF, 4.000€, **Open Science Ambassador.** Supporting the promotion and implementation of open science practices across research communities at Berlin Universities, with a focus on reproducibility, FAIR data, and fostering collaborative exchange.
- 10/2025 – 9/2027, Recipient, DFG, 219.000€, total: 895.500€, **SPP2311 - SimLivA - SIMulation supported LIVer Assessment for donor organs.** Developing computational models and simulation workflows to assess liver function in donor organs, advancing predictive tools for transplantation and improving clinical decision-making.
- 04/2023 – 09/2023, Recipient, BMBF, 1.500€, **X-Student Research Group - Physiologically based digital twins for the treatment of hypertension with ACE inhibitors and diuretics.** Project-based teaching initiative on digital twins in pharmacology, focusing on physiologically based pharmacokinetic (PBPK) modeling to optimize hypertension therapy with ACE inhibitors and diuretics.
- 03/2023 – 12/2026, Recipient, BMBF, 311.000€, total: 1530.550€, **ATLAS - AI and Simulation for Tumor Liver ASessment.** Developing AI-driven and simulation-based methods to improve the assessment of liver tumors, integrating computational modeling with clinical data to support personalized diagnosis and treatment planning.
- 10/2022 – 03/2023, Recipient, BMBF, 1.500€, **X-Student Research Group - Physiologically based modeling of drugs: ACE inhibitors in the treatment of high blood pressure.** Project-based teaching initiative on physiologically based pharmacokinetic (PBPK) modeling, focusing on optimizing the use of ACE inhibitors in hypertension therapy.
- 07/2021 – 11/2025, Recipient, DFG, 425.000€, total: 3825.000€, **FOR5151 - QuaLiPerF - Quantifying Liver Perfusion-Function Relationship in Complex Resection – A Systems Medicine Approach.** Advancing systems medicine by developing computational models to quantify the relationship between liver perfusion and function in complex surgical resections, supporting risk assessment and personalized treatment planning.
- 01/2020 – 12/2023, Co-Investigator, DFG, total: 658.942€, **SPP2311 - SimLivA - SIMulation supported LIVer Assessment for donor organs.** Collaborative project developing simulation approaches to assess donor liver function, aiming to improve transplantation outcomes through predictive computational modeling.

- 06/2020 – 05/2021, Recipient, EOSC-life, 5.000€, total: 25.000€, **EOSC-Life - Reproducible simulation studies targeting COVID-19.** Developed reproducible computational simulation studies to investigate COVID-19 dynamics, integrating standardized workflows and open science practices for rapid and transparent research dissemination.
- 06/2016 – 05/2022, Recipient, BMBF, 723.000€, **LiSyM - Systems Medicine of the Liver - Junior group leader - Multi-scale models for the personalized evaluation of liver function.** Led a junior research group developing multi-scale computational models to assess and predict liver function, advancing personalized medicine approaches and translational applications in hepatology.

Academic Distinctions

Habilitation Completion Scholarship — Humboldt-Universität zu Berlin

Scholarship supporting the completion of the Habilitation. — 11/2025 - 05/2026

- Awarded for the successful finalization of the Habilitation at Humboldt-Universität zu Berlin.

Michael Stifel Prize — Friedrich Schiller University Jena

Awarded for outstanding interdisciplinary and data-driven research achievements. — 2023

- Recognized for contributions advancing computational modeling and systems medicine.

Google Summer of Code Scholarship — Google / Open Source Initiative

Scholarship to develop SBML tools as part of the Google Summer of Code program. — 2015

- Advanced open-source infrastructure for systems biology and reproducible modeling.

Scholarship Studienstiftung des Deutschen Volkes — Studienstiftung des Deutschen Volkes

Awarded a prestigious scholarship from the German National Academic Foundation. — 2005

- Selected among top students in Germany for academic excellence and potential.

Teaching

Digital Teaching Project

Course	Description
DHPE-PKPD – An Interactive and Open Course on Pharmacokinetic (PK) and Pharmacodynamic (PD) Modeling Large-scale digital teaching project DHPE (80 AE) Charité – Universitätsmedizin Berlin 10/2024 – 07/2025	• Designed and implemented an innovative interactive course integrating theory, simulation practice, and reflection. • Leveraged open-source tools (Jupyter, GitHub, Quarto) to create reproducible, FAIR-compliant teaching materials. • Applied real-world pharmacology case studies and interactive simulations to foster research-oriented learning. • Pioneered blended learning approaches, combining digital resources with in-person tutorials. • Released all course materials openly, promoting accessibility, transparency, and reusability. • Received highly positive student evaluations for interactivity, structure, and practical relevance.

Research Project–Based Teaching

Course	Description
X-Student Research Group: Digital Twins in Action – Optimizing Direct Oral Anticoagulant Use Berlin University Alliance (BUA) 10/2025 – 03/2026 (2 SWS)	• Leading students in developing a digital twin of apixaban to simulate pharmacokinetics and clinical outcomes. • Combining lectures, tutorials, and hands-on modeling for experiential learning. • Enabling participants to explore personalized dosing strategies and safety aspects in anticoagulant therapy.
X-Student Research Group: Physiologically Based Digital Twins for Hypertension Therapy	• Supervised interdisciplinary teams modeling hydrochlorothiazide and lisinopril using PBPK methods. • Trained students in absorption, distribution, metabolism, and excretion (ADME) processes.

Course	Description
Berlin University Alliance (BUA) 04/2023 – 09/2023 (2 SWS)	• Fostered collaboration between STEM and medical students in clinically relevant pharmacology projects.
X-Student Research Group: Physiologically Based Modeling of ACE Inhibitors Berlin University Alliance (BUA) 10/2022 – 03/2023 (2 SWS)	• Guided students in developing PBPK models for lisinopril and ramipril. • Integrated lectures, tutorials, and project-based work to link theory with practice. • Strengthened student competencies in computational pharmacokinetics and systems medicine.

Lecture and Course–Based Teaching

Course	Description
Theory, Tools and Methods in Biology – Pharmacokinetic/Pharmacodynamic Modeling Humboldt-Universität zu Berlin (HU) 04/2025 – 09/2025 (2 SWS) 04/2024 – 09/2024 (2 SWS) 04/2023 – 09/2023 (2 SWS)	• Led pharmacokinetic modeling submodule (DHPE-PKPD) • Designed and delivered lectures on drug distribution, clearance, elimination, PBPK modeling, and PD • Integrated clinical applications and drug–drug interaction case studies • Developed submodule as part of the DHPE program
Important Models of Quantitative Biology from the Literature Humboldt-Universität zu Berlin (HU) 10/2021 – 03/2022 (2 SWS) 10/2020 – 03/2021 (2 SWS) 10/2019 – 03/2020 (2 SWS) 10/2018 – 03/2019 (2 SWS)	• Taught core methods for Boolean and ODE modeling in metabolism and signaling • Supervised student implementation and critical analysis of published models • Fostered analytical skills through structured discussion and group projects
Models of Cellular Processes Humboldt-Universität zu Berlin (HU) 10/2019 – 03/2020 (2 SWS)	• Introduced constraint-based, Boolean, and ODE models of cellular processes • Taught parameter optimization, stochastic modeling, and sensitivity analysis • Applied methods to real-world biological data examples

Teaching Qualifications

Digital Health Professions Educator (DHPE) — Charité – Universitätsmedizin Berlin

Certificate in Digital Teaching for Health Professions (200 AE) — 2024 – 2025

- Completed faculty development program focused on digital teaching and innovation in healthcare education.
- Designed and implemented future-oriented teaching scenarios, integrating blended and digital learning.
- Enhanced institutional teaching capacity through innovative, research-based learning formats.
- Applied competencies to improve curriculum development and digitally supported learning in health sciences.

AI Teaching Certificate (KI-Lehrzertifikat) — HU Berlin, KI-Campus 2.0 & AI Skills Initiative

Higher education didactics qualification on Artificial Intelligence (100/150 AE) — 2025 – 2026

- Qualification focused on integrating AI concepts and learning formats into university curricula.
- Training aligned with German Society for Higher Education Didactics (DGHD) standards.
- Emphasis on foundational AI literacy, didactic design, and sustainable curricular implementation.

List of Publications

Publications

1. **SBML Level 3 Package: Flux Balance Constraints version 3**. B. G. Olivier, F. T. Bergmann, S. Keating, and M. König; J Integr Bioinform. 2026; 20260006, 10.1515/jib-2026-0006, IF: **1.5**, Last author
2. **The Parameter Estimation tables (PEtab) format version 2**. D. Pathirana, F. Fröhlich, S. Persson, F. T. Bergmann, M. König, P. F. Lang, S. Bang, S. Kemmer, J. Hasenauer, and D. Weindl; J Integr Bioinform. , 10.1515/jib-2026-0004, IF: **1.5**

3. **EFECT enables reliable and reproducible stochastic simulations across disciplines**. T. J. Sego, M. König, L. L. Fonseca, D. Pathirana, F. T. Bergmann, H. E. Grecco, M. Silberberg, S. Ray, B. Fain, A. C. Knapp, K. Tiwari, H. Hermjakob, H. M. Sauro, J. A. Glazier, R. C. Laubenbacher, and R. S. Malik-Sheriff; npj Complex (2026), 10.1038/s44260-026-00094-y, IF: **1.6**
4. **Reproducibility of a Physiologically Based Pharmacokinetic and Pharmacodynamic (PBPK/PD) Model of Dapagliflozin**. M. Elias, M. Myshkina, N. Nemitz, and M. König; Physiome. May 13, 2026, 10.36903/physiome.31368556, Last author
5. **Automated Segmentation of Hepatic Vessels and Lobules in Whole-Slide Images Using U-Net Models**. M. Bafna, M. König, S. Saalfeld, V. Moulisova, V. Liska, U. Dahmen, and M. Albadry; Front. Bioinform. 6:1713736, 10.3389/fbinf.2026.1713736, IF: **3.9**
6. **From FAIR to CURE: Guidelines for Computational Models of Biological Systems**. H. M. Sauro, E. Agmon, M. L. Blinov, J. H. Gennari, J. Hellerstein, A. Heydarabadipour, P. Hunter, B. E. Jardine, E. May, D. P. Nickerson, L. P. Smith, G. D. Bader, F. Bergmann, P. M. Boyle, A. Drager, J. R. Faeder, S. Feng, J. Freire, F. Frohlich, J. A. Glazier, T. E. Gorochowski, T. Helikar, S. Hoops, P. Imoukhuede, S. M. Keating, M. König, R. Laubenbacher, L. M. Loew, C. F. Lopez, W. W. Lytton, A. McCulloch, P. Mendes, C. J. Myers, J. G. Myers, L. Mulugeta, A. Niarakis, D. D. van Niekerk, B. G. Olivier, A. A. Patrie, E. M. Quardokus, N. Radde, J. M. Rohwer, S. Sahle, J. C. Schaff, T. Sego, J. Shin, J. L. Snoep, R. Vadigepalli, H. S. Wiley, D. Waltemath, and I. Moraru; npj Syst Biol Appl (2026), 10.1038/s41540-026-00651-0, IF: **3.6**
7. **A Physiologically Based Pharmacokinetic and Pharmacodynamic (PBPK/PD) Model of Dapagliflozin in Type 2 Diabetes Mellitus: The Effect of Dosing, Hepatorenal Impairment, and Food**. N. Nemitz, M. Elias, and M. König; Pharmaceutics 2026, 18(3), 287; https://doi.org/10.3390/pharmaceutics 18030287, 10.3390/pharmaceutics18030287, IF: **6.9**, Last author
8. **A Digital Twin of the Angiotensin II Receptor Blocker Losartan: Physiologically Based Modeling of Blood Pressure Regulation**. E. Tensil, M. Myshkina, and M. König; Pharmaceutics 2026, 18(2), 262; https://doi.org/10.3390/pharmaceutics18020262, 10.3390/pharmaceutics18020262, IF: **6.9**, Last author
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10. **A Digital Twin of Glimepiride for Personalized and Stratified Diabetes Treatment**. M. Elias and M. König; Front. Pharmacol. 16:1686415., 10.3389/fphar.2025.1686415, IF: **4.8**, Last author
11. **Reproducibility of a Digital Twin of Glimepiride for Personalized and Stratified Diabetes Treatment**. M. Elias and M. König; Physiome. October 20, 2025, 10.36903/physiome.28379193, Last author
12. **Inflammation and autophagy in peripheral nerves of rodent models with metabolic syndrome and type 2 diabetes mellitus**. P. Baum, T. Ebert, N. Klöting, S. Krupka, M. König, S. Paeschke, P. Stock, M. Bulc, M. Blüher, K. Palus, M. Nowicki, and J. Kosacka; Neurosci Res. 2025 Apr 17:S0168-0102(25)00070-7, 10.1016/j.neures.2025.04.002, IF: **2.4**
13. **Anti-Endoglin monoclonal antibody prevents the progression of liver sinusoidal endothelial inflammation and fibrosis in MASH**. S. Eissazadeh, P. Fikrova, J. U. Rathouska, I. Nemeckova, K. Tripska, M. Vasinova, R. Havelek, S. Mohammadi, I. C. Igreja Sa, C. Theuer, M. König, S. Micuda, and P. Nachtigal; Life Sci. 2025 Jan 29:123428, 10.1016/j.lfs.2025.123428, IF: **5.2**
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15. **Cross-Species Variability in Lobular Geometry and Cytochrome P450 Hepatic Zonation: Insights into CYP1A2, CYP2E1, CYP2D6 and CYP3A4**. M. Albadry, J. Kuettner, J. Grzegorzewski, O. Dirsch, E. Kindler, R. Klopfleisch, V. Liska, V. Moulisova, S. Nickel, R. Palek, J. Rosendorf, S. Saalfeld, U. Settmacher, H. Tautenhahn, M. KönigΔ, and U. DahmenΔ (Δ equal contribution); Front Pharmacol. 2024 May 16;15:1404938, 10.3389/fphar.2024.1404938, IF: **5.4**, Last Equal author
16. **Bayesian modelling of time series data (BayModTS)-a FAIR workflow to process sparse and highly variable data**. S. Höpfl, M. Albadry, U. Dahmen, K. H. Herrmann, E. M. Kindler, M. König, J. R. Reichenbach,

H. M. Tautenhahn, W. Wei, W. T. Zhao, and N. E. Radde; Bioinformatics. 2024 May 2;40(5):btae312, 10.1093/bioinformatics/btae312, IF: **6.0**

17. **The simulation experiment description markup language (SED-ML): language specification for level 1 version 5**. L. P. Smith, F. T. Bergmann, A. Garny, T. Helikar, J. Karr, D. Nickerson, H. Sauro, D. Waltemath, and M. König; J Integr Bioinform. 2024 Apr 15, 10.1515/jib-2024-0008, IF: **1.5**, Last author
18. **Quantifying fat zonation in liver lobules: an integrated multiscale in silico model combining disturbed microperfusion and fat metabolism via a continuum biomechanical bi-scale, tri-phasic approach**. L. Lambers, N. Waschinsky, J. Schleicher, M. König, H. Tautenhahn, M. Albadry, U. Dahmen, and T. Ricken; Biomech Model Mechanobiol. 2024 Feb 25., 10.1007/s10237-023-01797-0, IF: **3.1**
19. **Eleven Strategies for Making Reproducible Research and Open Science Training the Norm at Research Institutions**. F. E. Kohrs, S. Auer, A. Bannach-Brown, S. Fiedler, T. L. Haven, V. Heise, C. Holman, F. Azevedo, R. Bernard, A. Bleier, N. Bössel, B. P. Cahill, L. J. Castro, A. Ehrenhofer, K. Eichel, M. Frank, C. Frick, M. Friese, A. Gärtner, K. Gierend, D. J. Grüning, L. Hahn, M. Hülsemann, M. Ihle, S. Illius, L. König, M. König, L. Kulke, A. Kutlin, F. Lammers, D. M. A. Mehler, C. Miehl, A. Müller-Alcazar, C. Neuendorf, H. Niemeyer, F. Pargent, A. Peikert, C. U. Pfeuffer, R. Reinecke, J. P. Röer, J. L. Rohmann, A. Sánchez-Tójar, S. Scherbaum, E. Sixtus, L. Spitzer, V. M. Straßburger, M. Weber, C. J. Whitmire, J. Zerna, D. Zorbek, P. Zumstein, and T. L. Weissgerber; eLife (2023) 12:e89736., 10.7554/eLife.89736, IF: **6.4**
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23. **Cytochrome p450 enzymes in periportal steatosis: Modulation of drug metabolizing activity but not of pericentral expression pattern**. M. Albadry, S. Höpfl, N. Ehteshamzad, M. König, M. Böttcher, J. Neumann, A. Lupp, O. Dirsch, N. Radde, B. Christ, M. Christ, L. Schwen, H. Laue, R. Klopfleisch, and U. Dahmen; Sci Rep. 2022 Dec 17;12(1):21825, 10.1038/s41598-022-26483-6, IF: **3.8**
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25. **BioSimulators: a central registry of simulation engines and services for recommending specific tools**. B. Shaikh, L. P. Smith, D. Vasilescu, G. Marupilla, M. Wilson, E. Agmon, H. Agnew, S. S. Andrews, A. Anwar, M. E. Beber, F. T. Bergmann, D. Brooks, L. Brusch, L. Calzone, K. Choi, J. Cooper, J. Detloff, B. Drawert, M. Dumontier, G. B. Ermentrout, J. R. Faeder, A. P. Freiburger, F. Fröhlich, A. Funahashi, A. Garny, J. H. Gennari, P. Gleeson, A. Goelzer, Z. Haiman, J. Hasenauer, J. L. Hellerstein, H. Hermjakob, S. Hoops, J. C. Ison, D. Jahn, H. V. Jakubowski, R. Jordan, M. Kalaš, M. König, W. Liebermeister, R. S. M. Sheriff, S. Mandal, R. McDougal, J. K. Medley, P. Mendes, R. Müller, C. J. Myers, A. Naldi, T. V. N. Nguyen, D. P. Nickerson, B. G. Olivier, D. Patoliya, L. Paulevé, L. R. Petzold, A. Priya, A. K. Rampadarath, J. M. Rohwer, A. S. Saglam, D. Singh, A. Sinha, J. Snoep, H. Sorby, R. Spangler, J. Starruß, P. J. Thomas, D. van Niekerk, D. Weindl, F. Zhang, A. Zhukova, A. P. Goldberg, J. C. Schaff, M. L. Blinov, H. M. Sauro, I. I. Moraru, and J. R. Karr; Nucleic Acids Res. 2022 May 7;50(W1):W108–14, 10.1093/nar/gkac331, IF: **16.7**
26. **Pharmacokinetics of caffeine: A systematic analysis of reported data for application in metabolic phenotyping and liver function testing**. J. Grzegorzewski, F. Bartsch, A. Köller, and M. König; Frontiers in Pharmacology 2022, Vol12, 10.3389/fphar.2021.752826, IF: **4.4**, Last author
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33. **OMEX Metadata Specification Version 1.2**. J. Gennari, M. König, G. Misirli, M. Neal, D. Nickerson, and D. Waltemath; J Integr Bioinform. 2021 Oct 20;18(3):20210020, 10.1515/jib-2021-0020, IF: **1.5**

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